ROCSCIENCE

SLIDE - An Interactive Slope Stability Program
Rocscience Inc.
Date Created: 12/12/2016
Software Version: 9.041

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Slide2 Analysis Information

SLIDE - An Interactive Slope Stability Program

Project Summary

Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:00.788s
Author:	Rocscience Inc.
Company:	Rocscience Inc.
Date Created:	12/12/2016

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Standard
Failure Direction:	Left to Right

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
	Spencer
Number of slices:	30
Tolerance:	0.0001
Maximum number of iterations:	50
Check malpha < 0.2:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Ru Coefficient
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	rand

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	10
Circles per division:	10
Number of iterations:	10
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Materials

Material 1	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	18 kN/m3
Cohesion	10.8 kPa
Phi	40 °
Ru Value	0.5

Global Minimums

Method: bishop simplified

FS	1.001250
Center:	69.472, 90.150
Radius:	80.152
Left Slip Surface Endpoint:	6.948, 40.000
Right Slip Surface Endpoint:	69.999, 10.000
Resisting Moment:	252759 kN-m
Driving Moment:	252443 kN-m
Total Slice Area:	421.748 m2
Surface Horizontal Width:	63.0515 m
Surface Average Height:	6.68894 m

Method: spencer

FS	1.018880
Center:	69.621, 90.016
Radius:	80.017
Left Slip Surface Endpoint:	7.163, 40.000
Right Slip Surface Endpoint:	69.999, 10.000
Resisting Moment:	253635 kN-m
Driving Moment:	248936 kN-m
Resisting Horizontal Force:	2879.04 kN
Driving Horizontal Force:	2825.69 kN
Total Slice Area:	415.971 m2
Surface Horizontal Width:	62.8365 m
Surface Average Height:	6.61989 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	2523
Number of Invalid Surfaces:	0

Method: spencer

Number of Valid Surfaces:	2341
Number of Invalid Surfaces:	182

Error Codes

Error Code -108 reported for 22 surfaces
Error Code -111 reported for 160 surfaces

Error Code Descriptions

The following errors were encountered during the computation:
-108 = Total driving moment or total driving force < 0.1. This is to limit the calculation of extremely high safety factors if the driving force is very small (0.1 is an arbitrary number).
-111 = Safety factor equation did not converge

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 1.00125

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	2.10172	47.5394	-50.0959	Material 1	10.8	40	10.1214	10.1341	10.516	11.3097	-0.793676	22.6193	11.3096
2	2.10172	132.968	-47.8063	Material 1	10.8	40	19.3806	19.4048	41.8882	31.6333	10.2549	63.2667	31.6334
3	2.10172	181.749	-45.6138	Material 1	10.8	40	25.3325	25.3642	60.5951	43.2381	17.357	86.4762	43.2381
4	2.10172	220.341	-43.5041	Material 1	10.8	40	30.4761	30.5142	75.9137	52.4193	23.4944	104.839	52.4193
5	2.10172	253.447	-41.4658	Material 1	10.8	40	35.2285	35.2725	89.4604	60.2953	29.1651	120.59	60.2951
6	2.10172	281.582	-39.4899	Material 1	10.8	40	39.5876	39.6371	101.355	66.9885	34.3668	133.977	66.9886
7	2.10172	305.167	-37.5687	Material 1	10.8	40	43.5524	43.6068	111.697	72.5995	39.0977	145.199	72.5996
8	2.10172	324.556	-35.6959	Material 1	10.8	40	47.1218	47.1807	120.569	77.212	43.357	154.424	77.2123
9	2.10172	340.043	-33.8661	Material 1	10.8	40	50.2948	50.3577	128.04	80.8965	47.1431	161.793	80.8967
10	2.10172	351.882	-32.0748	Material 1	10.8	40	53.0702	53.1365	134.168	83.7131	50.4547	167.426	83.7131
11	2.10172	360.289	-30.318	Material 1	10.8	40	55.446	55.5153	139.003	85.713	53.2897	171.426	85.7131
12	2.10172	365.45	-28.5922	Material 1	10.8	40	57.4196	57.4914	142.585	86.9408	55.6446	173.882	86.9407
13	2.10172	367.527	-26.8944	Material 1	10.8	40	58.9879	59.0616	144.951	87.4349	57.5162	174.87	87.4351
14	2.10172	366.662	-25.2216	Material 1	10.8	40	60.1465	60.2217	146.128	87.2291	58.8985	174.458	87.229
15	2.10172	362.977	-23.5716	Material 1	10.8	40	60.8904	60.9665	146.139	86.3526	59.7861	172.705	86.3526
16	2.10172	356.583	-21.9421	Material 1	10.8	40	61.2133	61.2898	145.003	84.8313	60.1716	169.663	84.8315
17	2.10172	347.574	-20.3311	Material 1	10.8	40	61.108	61.1844	142.734	82.688	60.0459	165.376	82.6882
18	2.10172	336.034	-18.7367	Material 1	10.8	40	60.5659	60.6416	139.342	79.9427	59.3988	159.885	79.9425
19	2.10172	322.037	-17.1573	Material 1	10.8	40	59.5773	59.6518	134.832	76.6129	58.2192	153.226	76.6128
20	2.10172	305.649	-15.5911	Material 1	10.8	40	58.1309	58.2036	129.208	72.7141	56.4936	145.428	72.7144
21	2.10172	286.927	-14.0369	Material 1	10.8	40	56.2139	56.2842	122.466	68.26	54.206	136.52	68.2601
22	2.10172	265.919	-12.4931	Material 1	10.8	40	53.8116	53.8789	114.602	63.2624	51.3394	126.525	63.2624
23	2.10172	242.67	-10.9585	Material 1	10.8	40	50.9077	50.9713	105.606	57.7315	47.8743	115.463	57.7315
24	2.10172	217.217	-9.43179	Material 1	10.8	40	47.4833	47.5427	95.4645	51.6761	43.7884	103.352	51.6763
25	2.10172	189.591	-7.91184	Material 1	10.8	40	43.5176	43.572	84.1598	45.1039	39.0559	90.2075	45.1036
26	2.10172	159.819	-6.39748	Material 1	10.8	40	38.9867	39.0354	71.6705	38.0209	33.6496	76.0418	38.0209
27	2.10172	127.921	-4.88759	Material 1	10.8	40	33.8637	33.906	57.969	30.4324	27.5366	60.8647	30.4323
28	2.10172	93.9139	-3.3811	Material 1	10.8	40	28.1183	28.1534	43.0231	22.3422	20.6809	44.6843	22.3421
29	2.10172	57.8105	-1.87695	Material 1	10.8	40	21.716	21.7431	26.7946	13.7532	13.0414	27.5063	13.7531
30	2.10172	19.6179	-0.374093	Material 1	10.8	40	14.6178	14.6361	9.23877	4.66711	4.57166	9.33422	4.66711

Global Minimum Query (spencer) - Safety Factor: 1.01888

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	2.09455	47.2936	-50.1424	Material 1	10.8	40	9.57362	9.75437	10.0436	11.2897	-1.24613	21.5107	10.221
2	2.09455	129.992	-47.8545	Material 1	10.8	40	14.2542	14.5234	35.4683	31.031	4.43728	51.2186	20.1876
3	2.09455	177.028	-45.6638	Material 1	10.8	40	17.946	18.2848	51.1794	42.2592	8.9202	69.5462	27.287
4	2.09455	215.496	-43.5559	Material 1	10.8	40	21.6669	22.076	64.8801	51.4421	13.438	85.4815	34.0394
5	2.09455	248.511	-41.5195	Material 1	10.8	40	25.4723	25.9532	77.3821	59.3233	18.0588	99.9335	40.6102
6	2.09455	276.584	-39.5454	Material 1	10.8	40	29.3051	29.8584	88.7377	66.0248	22.7129	112.934	46.9092
7	2.09455	300.136	-37.626	Material 1	10.8	40	33.1171	33.7424	98.9889	71.647	27.3419	124.516	52.8694
8	2.09455	319.517	-35.755	Material 1	10.8	40	36.8663	37.5623	108.168	76.2735	31.8942	134.712	58.439
9	2.09455	335.022	-33.9271	Material 1	10.8	40	40.5145	41.2794	116.299	79.9748	36.3239	143.551	63.5763
10	2.09455	346.902	-32.1376	Material 1	10.8	40	44.0271	44.8583	123.4	82.8107	40.5892	151.058	68.2476
11	2.09455	355.372	-30.3827	Material 1	10.8	40	47.371	48.2654	129.482	84.8325	44.6497	157.255	72.4228
12	2.09455	360.617	-28.6588	Material 1	10.8	40	50.5143	51.468	134.551	86.0846	48.4661	162.159	76.0746
13	2.09455	362.799	-26.9627	Material 1	10.8	40	53.4253	54.434	138.606	86.6054	52.0007	165.784	79.1785
14	2.09455	362.058	-25.2919	Material 1	10.8	40	56.0718	57.1304	141.643	86.4285	55.2144	168.138	81.7098
15	2.09455	358.517	-23.6439	Material 1	10.8	40	58.4206	59.5236	143.65	85.5833	58.0664	169.226	83.643
16	2.09455	352.285	-22.0163	Material 1	10.8	40	60.4369	61.5779	144.61	84.0955	60.5149	169.049	84.953
17	2.09455	343.456	-20.4072	Material 1	10.8	40	62.0832	63.2553	144.502	81.988	62.5137	167.599	85.6112
18	2.09455	332.114	-18.8148	Material 1	10.8	40	63.3193	64.5148	143.295	79.2806	64.0146	164.869	85.5886
19	2.09455	318.334	-17.2374	Material 1	10.8	40	64.1011	65.3113	140.955	75.9909	64.9639	160.843	84.8524
20	2.09455	302.179	-15.6733	Material 1	10.8	40	64.3792	65.5947	137.436	72.1345	65.3017	155.5	83.3656
21	2.09455	283.706	-14.1212	Material 1	10.8	40	64.0988	65.309	132.686	67.7249	64.9615	148.812	81.0871
22	2.09455	262.966	-12.5795	Material 1	10.8	40	63.1977	64.3909	126.641	62.774	63.867	140.744	77.9696
23	2.09455	240.002	-11.0471	Material 1	10.8	40	61.605	62.7681	119.225	57.292	61.9333	131.253	73.9606
24	2.09455	214.849	-9.52258	Material 1	10.8	40	59.2389	60.3573	110.348	51.2877	59.06	120.285	68.9971
25	2.09455	187.541	-8.00488	Material 1	10.8	40	56.0044	57.0618	99.9017	44.7689	55.1328	107.777	63.0086
26	2.09455	158.103	-6.49282	Material 1	10.8	40	51.7907	52.7685	87.7575	37.7416	50.0159	93.6517	55.9101
27	2.09455	126.557	-4.98529	Material 1	10.8	40	46.4665	47.3438	73.7621	30.211	43.5511	77.8153	47.6043
28	2.09455	92.9187	-3.48122	Material 1	10.8	40	39.8759	40.6288	57.7296	22.1811	35.5485	60.1554	37.9743
29	2.09455	57.2012	-1.97954	Material 1	10.8	40	31.8315	32.4325	39.4354	13.6548	25.7806	40.5356	26.8808
30	2.09455	19.4119	-0.479227	Material 1	10.8	40	22.0921	22.5092	18.5884	4.6339	13.9545	18.7731	14.1392

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 1.00125

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	6.94775	40	0	0	0
2	9.04947	37.4867	5.15713	0	0
3	11.1512	35.1684	61.5372	0	0
4	13.2529	33.0211	138.407	0	0
5	15.3546	31.0264	225.783	0	0
6	17.4563	29.1692	317.889	0	0
7	19.5581	27.4373	410.224	0	0
8	21.6598	25.8206	499.272	0	0
9	23.7615	24.3106	582.295	0	0
10	25.8632	22.9001	657.188	0	0
11	27.9649	21.583	722.365	0	0
12	30.0666	20.3539	776.672	0	0
13	32.1684	19.2084	819.327	0	0
14	34.2701	18.1424	849.869	0	0
15	36.3718	17.1524	868.119	0	0
16	38.4735	16.2355	874.151	0	0
17	40.5752	15.3888	868.269	0	0
18	42.6769	14.61	850.99	0	0
19	44.7787	13.8971	823.034	0	0
20	46.8804	13.2483	785.308	0	0
21	48.9821	12.6618	738.908	0	0
22	51.0838	12.1364	685.113	0	0
23	53.1855	11.6707	625.383	0	0
24	55.2872	11.2637	561.366	0	0
25	57.389	10.9146	494.899	0	0
26	59.4907	10.6225	428.019	0	0
27	61.5924	10.3869	362.969	0	0
28	63.6941	10.2071	302.216	0	0
29	65.7958	10.083	248.462	0	0
30	67.8975	10.0141	204.666	0	0
31	69.9993	10.0004	0	0	0

Global Minimum Query (spencer) - Safety Factor: 1.01888

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	7.16277	40	0	0	0
2	9.25732	37.4912	5.14318	2.23625	23.4994
3	11.3519	35.1768	57.3717	24.9451	23.4993
4	13.4464	33.0331	129.49	56.3021	23.4993
5	15.541	31.0416	213.315	92.7492	23.4994
6	17.6355	29.1872	303.453	131.941	23.4993
7	19.7301	27.4578	395.529	171.975	23.4993
8	21.8246	25.8433	485.977	211.302	23.4993
9	23.9192	24.3351	571.883	248.654	23.4993
10	26.0137	22.9262	650.871	282.998	23.4994
11	28.1083	21.6104	721.018	313.498	23.4994
12	30.2028	20.3824	780.794	339.488	23.4993
13	32.2974	19.2376	829.009	360.452	23.4993
14	34.3919	18.1721	864.783	376.006	23.4993
15	36.4865	17.1824	887.515	385.89	23.4993
16	38.581	16.2654	896.865	389.956	23.4994
17	40.6756	15.4184	892.742	388.163	23.4993
18	42.7701	14.6392	875.298	380.578	23.4993
19	44.8647	13.9255	844.922	367.371	23.4993
20	46.9592	13.2757	802.249	348.817	23.4994
21	49.0538	12.688	748.162	325.3	23.4993
22	51.1483	12.161	683.809	297.319	23.4993
23	53.2429	11.6936	610.618	265.496	23.4993
24	55.3374	11.2847	530.325	230.584	23.4993
25	57.432	10.9333	445.006	193.488	23.4993
26	59.5265	10.6388	357.117	155.274	23.4993
27	61.6211	10.4004	269.548	117.199	23.4993
28	63.7156	10.2177	185.69	80.7375	23.4993
29	65.8102	10.0903	109.515	47.6171	23.4994
30	67.9047	10.0179	45.6912	19.8665	23.4994
31	69.9993	10.0004	0	0	0

Discharge Sections

Entity Information

External Boundary

X	Y
0	0
80	0
80	10
70	10
10	40
0	40